AI DRIVEN STORY TELLING WITH ART

水墨 *Traditional Chinese Ink Painting Style*

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PROBLEM

- **Modern bias**
  Diffusion models default to photorealistic, Western imagery.
- **Cultural drift**
  Ink-wash conventions are inconsistently preserved across generations.
- **Disjoint pipeline**
  Story writing and image synthesis live in separate, uncoordinated steps.
- **Prompt mismatch**
  Free-text prompts misalign with the target visual grammar.

> *"Off-the-shelf models render what is plausible, not what is rooted in tradition."*

SOLUTION

3900 *Source Images Collected* → **2318** *Curated & captioned*

- **Data curation**
  Heuristic + manual filtering for stylistic purity.
- **Captioning**
  Automated tagging refined by human review.
- **LoRA fine-tuning**
  Stable Diffusion 1.5 adapted to ink-wash domain.
- **Scene-first prompting.**
  Compositional prompts before pixel synthesis.
- **Story model**
  Qwen 2.5–3B for narrative scaffolding.

RESULTS

- Consistent Ink-wash style across scenes
- Coherent - Multiscene story telling
- Improved Alignment between text and Images

SYSTEM ARCHITECTURE

USER INPUT (STORY IDEA) → Frontend (WEB UI) → Backend (ORCHESTRATOR) → LLM Story Gen. (QWEN 2.5 – 3B) → Scene Prompts (REFINED) → Diffusion + LoRA (SD 1.5 FINE TUNED) → Generated Images (INK-WASHED SCENES) → STORY BOOK (json + IMAGES)

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HOW IT WORKS

I > Story Input
- User provides story idea
- LLM generates scenes

II > Scene Processing
- Scenes converted into structured prompts
- Prompt refinement applied

III > Image Generation
- Diffusion + LoRA generates images
- Outputs combined into storybook

IMPACT

- AI + cultural heritage integration
- Automated storytelling system
- Educational and creative applications

CHALLENGES

- Character consistency across scenes
- Limited rendering of detailed humans
- Prompt length constraints

FUTURE WORK

- Improve character consistency across the screens
- Expand to multiple art styles and cultures
- Deploy as web application for web access and interactive demos

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